

All of Statistics
Chapter 1: Probability
Solutions to Exercises

Jérémie Mabiala

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Exercise 1 Must be addedd.

Exercice 2 Page 6.

Looking at the Venn diagram fig.1, we have :

$$A \cup B = (A \setminus B) \cup (B \setminus A) \cup (A \cap B)$$

$$A = (A \setminus B) \cup A \cap B$$

$$B = (B \setminus A) \cup (A \cap B)$$

Because, the probability of a union of disjoint events is the sum of the probabilities of each event, we have :

$$P(A \cup B) = P(A \setminus B) + P(B \setminus A) + P(A \cap B)$$

$$P(A) = P(A \setminus B) + P(A \cap B)$$

$$P(B) = P(B \setminus A) + P(A \cap B)$$

Since $P(A \cap B) = P(\emptyset) = 0$, we have :

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

which concludes the proof. $\square$

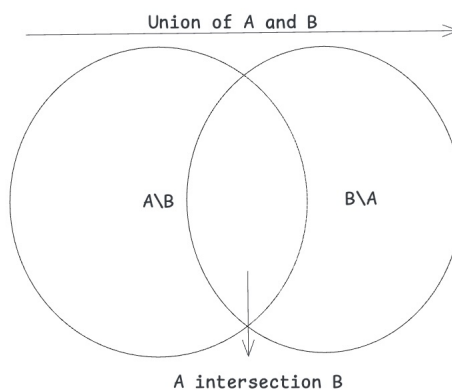


Figure 1: Venn diagram for A and B

Exercise 3 (a) 1) We have to show that for all $n \in \mathbb{N}$, $B_{n+1} \subset B_n$. Let $n \in \mathbb{N}$. By definition, we have

$$B_n = \bigcup_{i=n}^{\infty} A_i, \quad B_{n+1} = \bigcup_{i=n+1}^{\infty} A_i$$

Now, expand B_n as follows:

$$B_n = A_n \cup \bigcup_{i=n+1}^{\infty} A_i = A_n \cup B_{n+1}$$

From this, it is clear that $B_{n+1} \subset B_n$.

2) We have to show that $C_n \subset C_{n+1}$ for all $n \in \mathbb{N}$. Let $n \in \mathbb{N}$. Let $T_i = A_i^c$ for all $i \in \mathbb{N}$, such that $Z_n = \Omega \setminus C_n = \bigcup_{i=n}^{\infty} T_i$. From part (a) 1), we know that $Z_{n+1} \subset Z_n$. Taking the complement on both sides, we get $C_n \subset C_{n+1}$.

Exercise 4 Let's recall this first. If $\{A_i, i \in I\}$ is a collection of sets indexed by a set I , then the intersection and union over I is defined as

$$\begin{aligned} w \in \bigcap_{i \in I} A_i &\iff \forall i \in I, w \in A_i \\ w \in \bigcup_{i \in I} A_i &\iff \exists i \in I, w \in A_i \end{aligned}$$

Now,

$$\begin{aligned} \left(\bigcup_{i \in I} A_i \right)^c &= \{w : w \notin \bigcup_{i \in I} A_i\} \\ &= \{w : \forall i \in I, w \notin A_i\} \\ &= \{w : w \in A_i^c, \forall i \in I\} \\ &= \bigcap_{i \in I} A_i^c \end{aligned}$$

and

$$\begin{aligned} \left(\bigcap_{i \in I} A_i \right)^c &= \{w : w \notin \bigcap_{i \in I} A_i\} \\ &= \{w : \exists i \in I, w \notin A_i\} \\ &= \{w : \exists i \in I, w \in A_i^c\} \\ &= \bigcup_{i \in I} A_i^c \end{aligned}$$